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⑥3 Regular Grammar

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Noon Chomsky gave a Mathematical model for Grammar which is effective for writing computer languages.

Grammar Type	Grammar Accepted	Language Accepted
<u>Type-3</u>	<u>Regular Grammar</u>	<u>Regular language</u>

Finite State Automaton

<u>Type-2</u>	<u>Context Free grammar</u>	<u>Context Free Language</u>
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Pushdown Automata

Type-1	Context Sensitive Grammar	Context sensitive
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Language

Linear Bounded Automaton

Type-0	Unrestricted Grammar	Recursively Enumerable Language
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Turing machine



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Grammar:

A Grammar 'G' can be formally described using 4 tuples as $G = (V, T, S, P)$ where,

- V = Set of Variables or Non-Terminal Symbols
- T = Set of Terminal Symbols
- S = Start Symbol
- P = Production rules for Terminals and Non-Terminals

A production rule has the form $\alpha \rightarrow \beta$ where α and β are strings on $V \cup T$ and at least one symbol of α belongs to V

Example: $G = (\{S, A, B\}, \{a, b\}, S, \{S \rightarrow AB, A \rightarrow a, B \rightarrow b\})$

$\Rightarrow V = \{S, A, B\}, T = \{a, b\},$

$S = S, P = \{S \rightarrow AB, A \rightarrow a, B \rightarrow b\}$

Eg. $S \rightarrow AB$

$\rightarrow aB$

$\rightarrow ab$

Regular Grammar :

two types:

Right Linear Grammar:

A grammar is said to be Right Linear if all productions are of the form

$$A \rightarrow xB$$

$$A \rightarrow x$$

where $A, B \in V$ and $x \in T$

eg, $S \rightarrow abS \mid b$ - Right Linear

Left Linear Grammar:

$\sim$ is said $\sim$ Left Linear $\sim$

$$A \rightarrow Bx$$

$$A \rightarrow x$$

where $A, B \in V$ and $x \in T$

eg, $S \rightarrow Sb \mid b$ $\Rightarrow$ Left Linear.

(64) Derivations from a Grammar

The set of all strings that can be derived from a Grammar is said to be the LANGUAGE generated from that Grammar

e.g.

$$G_1 = (\{S, A\}, \{a, b\}, S, \{S \rightarrow aAb, aA \rightarrow aaAb, A \rightarrow \epsilon\})$$

$$S \rightarrow \underline{a}Ab \quad [\text{by } S \rightarrow aAb]$$

$$\rightarrow aaAb \quad [\text{by } aA \rightarrow aaAb]$$

$$\rightarrow aa\underline{a}Ab \quad [\text{by } aA \rightarrow aaAb]$$

$$\rightarrow aaabbb \quad [\text{by } A \rightarrow \epsilon] \quad \text{LCG}_1 = a^3b^3$$

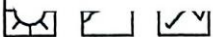
example 2:

$$G_2 = (\{S, A, B\}, \{a, b\}, S, \{S \rightarrow AB, A \rightarrow a, B \rightarrow b\})$$

$$S \rightarrow AB$$

$$\rightarrow ab \quad \text{only string.}$$

$$L(G_2) = \{ab\}$$



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Example 3.

$$G = (\{S, A, B\}, \{a, b\}, S, \{S \rightarrow AB, A \rightarrow aA|a, B \rightarrow bB|b\})$$

$$\textcircled{1} S \rightarrow AB$$

$$\rightarrow ab$$

$$\textcircled{2} S \rightarrow AB$$

$$\rightarrow aAbB$$

$$\rightarrow aabB$$

$$\textcircled{3} S \rightarrow AB$$

$$\rightarrow aAb$$

$$\rightarrow aab$$

$$\textcircled{4} S \rightarrow AB$$

$$\rightarrow abB$$

$$\rightarrow abb$$

$$L(G) = a^{\dagger}ab b^{\dagger}$$

$$= \{a^m b^n\}$$

$$= \{a^m b^n \mid m \geq 0 \text{ and } n \geq 0\}$$